Date	15/07/2026
Team ID	xxxxxx
Project Name	Smart Lender (Loan Approval Prediction)
Maximum Marks	10 Marks

Sample Project Documentation Report

Smart Lender (Loan Approval Prediction) Project Report

Project Summary: Smart Lender is an automated credit evaluation system. Fleshed out on a loan dataset, the system maps personal characteristics and financial indicators to a binary decision: Approved or Rejected. By automating the screening process, financial institutions reduce underwriting cycle times and eliminate subjective biases, guaranteeing a fairer and faster experience.

Objectives:

• Analyze historical applicant parameters (credit history, income, property area). • Balance target classes using SMOTE techniques. • Build and evaluate DT, RandomForest, KNN, and XGBoost classifiers. • Integrate the best model (Random Forest, 81.3% test accuracy) into an interactive Flask application.

Model Evaluation:

The RandomForest model was selected due to its high accuracy and resistance to overfitting. Evaluation details: • Accuracy: 81.3% on test set. • Precision: 79.5% for approval prediction. • Recall: 98.4% (minimizing false negatives for creditworthy applicants). • Serialization: Saved using joblib into models/model.pkl.

Architecture Flow:

User Form Input -> Data Mapping -> StandardScaler -> Random Forest Classifier -> HTML Result Render.

Team: CHAKALA YOGESH (Lead), Thippireddy Gari Keerthi, Mohammad Gouse Siddiq Maniyar, Madasu Pavan Kuma